Sigrid Trägenap

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Nationality: German

Education

11/2018 – present	PhD student, **International Max Planck Research School for Neural Circuits, Frankfurt**; thesis submitted 01/2026
06/2020 – 11/2024	Funded by the **Center for Multiscale Modelling, Frankfurt**
10/2016 – 10/2018	**MSc Biophysics** **Goethe University Frankfurt** Thesis: The role of network interactions in refining neural response properties in visual cortex, supervisor: Matthias Kaschube **Final grade: 1.0**
10/2013 – 11/2016	**BSc Biophysics** **Goethe University Frankfurt** Thesis: Development of coding properties in mammalian visual cortex, supervisor: Matthias Kaschube **Final grade: 1.3**
2009 – 2013	**Landesschule Pforta, Department of Natural Science**, Abitur (university entrance qualification), **Final grade: 1.0**

Publications

The developmental emergence of reliable cortical representations.
Sigrid Trägenap, David E. Whitney, David Fitzpatrick, Matthias Kaschube;
Nature Neuroscience; 2025; 28: 394–405; doi: https://doi.org/10.1038/s41593-024-01857-3

Development of coherent cortical responses reflects increased discriminability of feedforward inputs and their alignment with recurrent circuits.
Augusto Lempel, Sigrid Trägenap, Clara Tepohl, Matthias Kaschube, David Fitzpatrick;
Neuron; 2025; 113(22): 3834–3845.e8; doi: https://doi.org/10.1016/j.neuron.2025.08.014

Millimeter-scale selective amplification in the developing visual cortex.
Haleigh N. Mulholland, Sigrid Trägenap, Matthias Kaschube, Gordon B. Smith;
Co-first authors
bioRxiv (preprint); 2026; doi: https://doi.org/10.64898/2026.03.31.715687

Selected conference presentations

Oral	Rhine-Main Neuroscience Meeting (Oberwesel) 2026 *invited* Giersch Conference on Digital Twins 2023 Computational Neuroscience Next Generation Symposium Vision Science Society Meeting 2023, St. Pete Beach, Florida German Neuroscience Meeting 2023, symposium Society for Neuroscience Annual Meeting 2022, nanosymposium, San Diego Ferret brain meeting 2022, San Diego

	Bernstein conference, main conference 2022, Berlin CNS Annual Computational Neuroscience Meeting 2019, Barcelona
Poster	Bernstein Conference 2026 HWK Workshop “Cortex Structure and Function in Mouse, Monkey and Man”, Delmenhorst, 2026 10th European Visual Cortex and Cognition Meeting, Cortona Cosyne 2026, Lisbon Bernstein conference 2025, Frankfurt 3rd Meeting on Spontaneous Activity in Brain development, Alicante, 2024 Bernstein conference 2024, Frankfurt Cosyne 2024, Lisbon 9th European Vision and Cognition Meeting, Il Poggio, 2024 Cosyne 2023, Montreal 2nd Meeting on Spontaneous Activity in Brain development, Alicante, 2022 Cosyne 2021 (virtual) Bernstein conference 2020 (virtual) Cosyne 2019, Lisbon 1st Meeting on Spontaneous Activity in Brain development, Leiden, 2018 Biennial Meeting of the Rhine-Main Neuroscience Network, 2018 Bernstein conference 2018, Berlin Bernstein conference 2017, Göttingen

Research Experience

01/2019 – 03/2019	Rotating PhD student with **Pascal Fries, Ernst-Strüngmann-Institute, Frankfurt** Analysis of electrophysiological recordings; the role of α-band activity
10/2017 – 10/2018	Student assistant with **Matthias Kaschube, Frankfurt Institute for Advanced Studies**, analysis of spontaneous activity in visual cortex
11/2016 – 03/2017 and 05/2016 – 09/2016	Student assistant with **Gilles Laurent, Max Planck Institute for Brain Research Frankfurt**, cuttlefish project
05/2012 and 06/2011 – 07/2012	Intern, **Martin-Luther University Halle-Wittenberg, Prof. Binder,** macromolecular chemistry

Teaching Experience

2021 – 2024	**Theoretical Neuroscience I & II** Teaching assistant: guest lectures, coordination of exercise sheets and grading, content preparation
2018 – 2026	**Computational Neuroscience Seminar** Teaching assistant: paper selection, introduction to scientific writing, feedback on talks and written reports
2017 – 2018	**Machine Learning I** Teaching assistant
2015 – 2016	**Mathematics for physicists I** **Mathematics for scientists II** Undergraduate teaching assistant: weekly tutorials, grading of exercise sheets

Awards and Scholarships

07/2023	**Travel Award** from Organization of Computational Neuroscience
09/2022	**Travel Award** from Bernstein Network
06/2022	**Poster Award** (1st Prize), 2nd Meeting on Spontaneous Activity in Brain development "Experience drives the development of novel cortical sensory representation from endogenous networks"
07/2019	**Travel Award** from Organization of Computational Neuroscience
02/2019	Cosyne Presenters **Travel Award** (Conference for Computational Systems Neuroscience)
06/2018	**Poster Award** (3rd prize), 5th Biennial Meeting of the Rhine-Main Neuroscience Network "The role of network interactions in coordinating neuronal response properties across visual cortex"
09/2017 – 09/2018	**SMART*START*** training programme in computational neuroscience, funded by the Bernstein Network and the Volkswagen Foundation
04/2014 – 10/2018	Scholarship from the **German National Academic Foundation**
10/2013 – 04/2014	**Deutschlandstipendium** (scholarship for high-achieving students with social commitment)
2013	**DPG Abiturientenpreis**, German Physical Society award for the best physics students at secondary-school level
2013	**eLeMeNTe e.V. honours**, for outstanding results in mathematics and science competitions in the state of Saxony-Anhalt

Voluntary Activities

03/2019 – present	Co-organiser of the German Neuroscience Olympiad final in Frankfurt (for secondary-school students) Outreach and press contact, grading coordinator, computational tutorial
12/2018 – 09/2022	Student representative, Frankfurt Institute for Advanced Studies Liaison with the administration, student support, organisation of social events
10/2013 – 02/2018	Student representative (Fachschaft Physik), Goethe University Frankfurt Member of the faculty council and the examination committee, organisation of course evaluations
2015 – 2018	Night of Science, Goethe University Frankfurt: organising team (purchasing, conceptual development of booths), executive committee 2018

Skills

Language	German (native) English (fluent, CEFR C2) Latin (good reading knowledge)
IT	Python (advanced) MATLAB (good) LaTeX (very good) Word, Excel, PowerPoint (very good)